TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 2

5-MINUTE REVISION

Official Agricultural Statistics

How India measures crop area, yield and production, and what the Agricultural Census records.

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STAT 3202 · Chapter revision

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THE IDEA TO KEEP

Check the source, year, unit and geographical coverage.

REMEMBER

Official figures answer different questions: area, production, yield and prices are not interchangeable.

APPLY IT

If production is 60 tonnes from 20 hectares, yield is 3 tonnes per hectare. Compare like units and reference years.

1 minute: key idea · 3 minutes: methods & examples · 1 minute: test yourself

1 Who collects what

- Agriculture is a **state subject**, but agricultural statistics are on the **concurrent list**, so the system is **decentralised**.
- **State Agricultural Statistics Authorities (SASAs)** collect and compile at state level.
- Nationally, the **ES&E Division** (Economics, Statistics and Evaluation) of the Dept. of Agriculture & Farmers Welfare compiles crop production estimates.

Crop production

$$\text{Production} = \text{Area} \times \text{Average yield}$$

2 Area statistics: three systems

System	Where	Share*
Complete enumeration	Temporarily settled states with cadastral land records	~86%
Sample surveys	Permanently settled states with no village revenue agency, e.g. **Kerala**, Odisha, West Bengal	~9%
No developed system	Hill and north-eastern states, some UTs (personal assessment)	~5%

*Share of the country's reporting area, ICS report 2018–19.

3 Crop Cutting Experiment (CCE): steps

A small, randomly located plot is harvested and weighed to estimate yield. Yield surveys for principal crops run under the **General Crop Estimation Survey (GCES)**.

1. **Select the field:** it must be larger than the CCE plot.
2. **Locate the plot** from the **south-west corner** using random steps along the horizontal and vertical sides.
3. **Mark and harvest** plants rooted inside the pegged, string-marked plot; apply the boundary rule consistently.
4. **Thresh** on a clean mat and dry the grain.
5. **Winnow and weigh** : subtract the empty bag to get the net grain weight.

4 CCE numbers

Check a square plot

$$d = a\sqrt{2};\ 5\ \text{m plot} \Rightarrow 7.07\ \text{m}$$

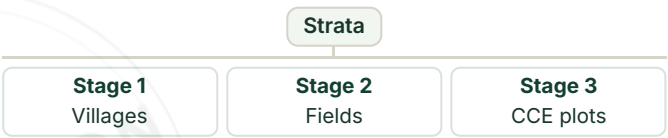
Plot yield (kg/ha)

$$\frac{10\,000\,w}{a}$$

w = net grain (kg), a = plot area (m²). 12.5 kg from 25 m² = 5000 kg/ha.

Plot shapes: 5×5 m, 10×10 m squares, 10×5 m rectangle; triangles (UP) and circles (WB) also documented.

5 GCES sampling design



A **multistage** design. Plots are chosen by the prescribed random procedure, never by convenience. One plot does not represent a whole region.

6 Production estimates released

Release	Usually	Covers
First Advance	September	Kharif
Second Advance	February	Kharif + Rabi
Third Advance	May	Updated area and yield
Final	Sept / Oct	Final estimates

Advance estimates exist because final figures come only after harvest. Quote any estimate with its year and release status.

7 Agricultural Census

What
Structure of agriculture: number and area of **operational holdings**, land use, cropping pattern, inputs, down to village level.

When
Started **1970–71**; **quinquennial** (every 5 years). 11th census: reference year **2021–22**, launched 28 July 2022.

How
Land-record states: number and area by **complete enumeration**; characteristics on a **sample basis**.

Reference period
The agricultural year, **July to June**.

! Common mistakes

- ✗ Kerala uses **sample surveys** for area, not complete enumeration.
- ✗ Yield = production ÷ **area**, never ÷ number of farmers.
- ✗ Don't mix up the current three advance estimates with older four-forecast schedules.

? Fill in the blanks

1. A district with 20 000 ha under paddy and an average yield of 4 t/ha produces ____ t. [80 000]
2. 12.5 kg of grain from a 25 m² CCE plot is a yield of ____ kg/ha. [5000]
3. Each diagonal of a 10 m × 10 m CCE plot should measure about ____ m. [14.14]
4. Kerala has no village revenue agency, so its crop area is estimated by ____. [sample surveys]
5. Kharif production first appears in the ____ Advance Estimates. [First]
6. The Agricultural Census is quinquennial, so the census after reference year 2021–22 refers to ____. [2026–27]